Electronics and Computer Technology

ECET 250E: Lab 04 Resistive DC Circuits **(**Kirchhoff's law and Voltage Divider)

Objectives:

- To confirm KVL and the idea of the voltage divider.
- To familiarize yourself with the design and use of Voltage Dividers.
- To use a potentiometer as a voltage divider.

INTRODUCTION

In this lab, you will familiarize yourself with *Kirchhoff's Voltage Law* and *voltage dividers* using the potentiometer.

Kirchhoff's Voltage Law and the Voltage Divider

Kirchhoff's Voltage Law (KVL) says that the algebraic sum of voltages around a mesh or loop of a circuit is zero. See **Figure 1**, the voltages around this loop are summed algebraically according to the polarity of the voltages across each component. The voltage source has a voltage "**rise**" of 12 V and each resistance "**drops**" a voltage (opposite in sign to the source) in proportion to the ratio of that resistance to the total resistance.

For example: the voltage across **R3** is:

$$V_{R3} = V_{CD} = \frac{R3}{R_{Total}} \times V_1 = \frac{2200\,\Omega}{11200\,\Omega} \times 12\,V = 2.36\,V$$

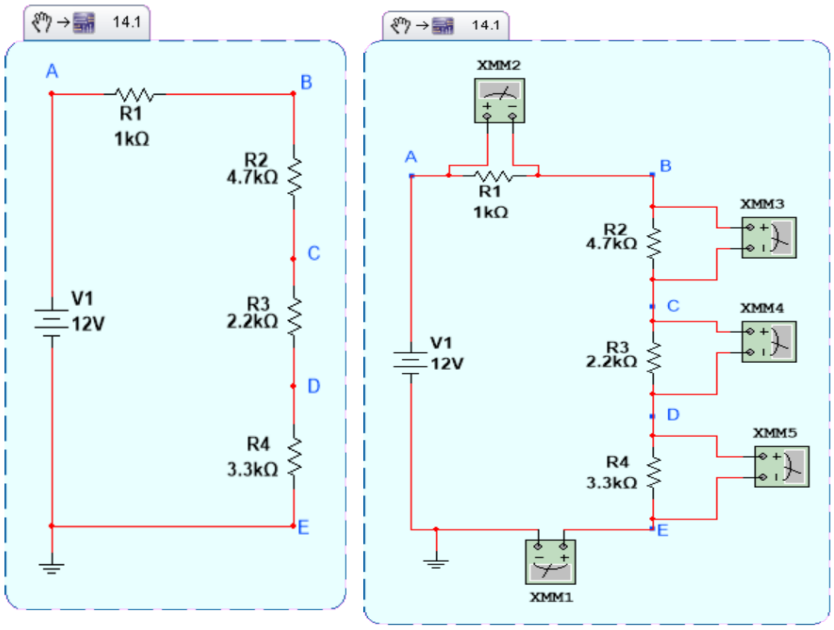
In this loop the sum of the voltages across the resistors will add up to the source voltage. We can say that the source voltage (12 V) is **divided** up between the resistors. In this sense we have a **Voltage Divider.**

If we could somehow combine all those resistances into one resistor and "tap off" that resistor at any point then we could realize any voltage from 0 – 12 V off that resistance. Such a device is called a **potentiometer**.

PROCEDURE

Part A: Voltage Dividers

Build the following circuit as shown in **Figure 1**. Measure the voltage across each resistor and the total current through the circuit. Record the data in **Table A**. Compare the measured values with the calculated values. The 'XMM1' is the ammeter (setup to measure current). (If you would like you can use the MultiSim to verify your answers too)



With a voltmeter measure the *voltage drops* between each point as specified in the **Table A**. Also measure the *total current* in the circuit.

RESULTS: Table A

MEASUREMENT	CALCULATED	MEASURED
V_{AB}		
V_{BC}		
V_{CD}	2.36V	
V_{DE}		
V_{AC}		
V_{DC}	-2.36V	
V_{EC}		
I_{TOTAL}	$I_T = \frac{V}{R_T}$	

- Confirm Ohm's Law (by measuring the circuit current and total circuit resistance). and explain differences (if any) between measured and calculated values.
- Compare $\mathbf{V_1}$ to the sum of all drop voltages (**KVL**).

Part B: Voltage Dividers and Potentiometers

Build the following circuits shown in **Figures 2, 3, and 4**. Measure the range of output voltage **between X and Y** (Max-Min) for each circuit (This means once Potentiometer will be at its full range (Max) and next time at minimum range (Min)). Record this data in **Table B**, on the following page. Compare the *measured* values with the *calculated* values.

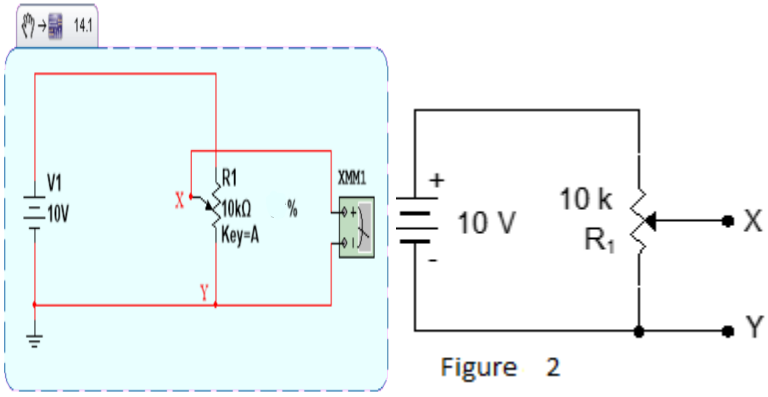


Figure 2.

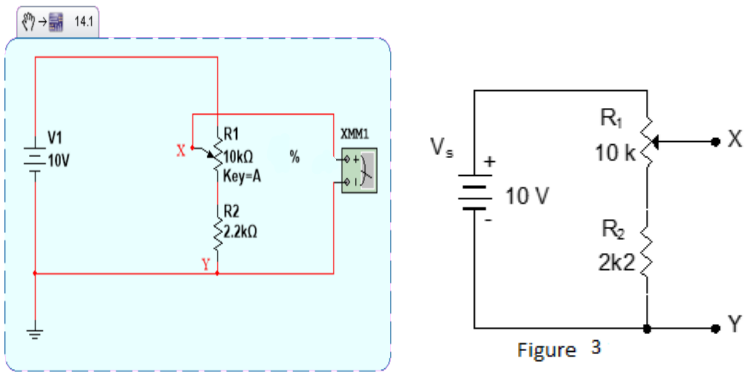


Figure 3.

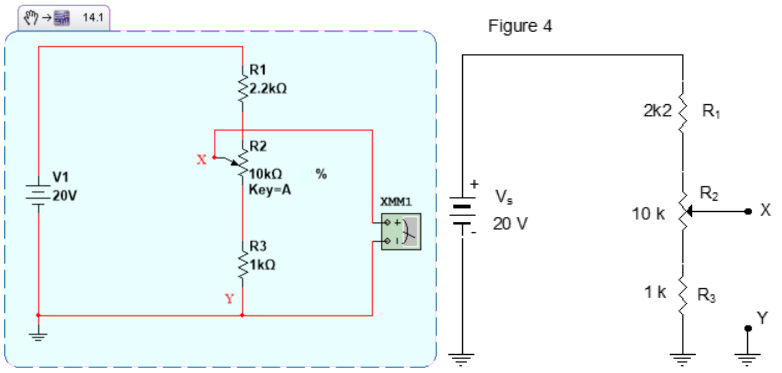


Figure 4.

RESULTS: Table B

Circuit	Calculated Range of V_{XY} (Max-Min)	Measured Range of V_{XY}(Max-Min)
Figure 2		
Figure 3		
Figure 4		

Discussion & Conclusion

a. Were the objectives met? Use your data to show that the lab objectives were met.
b. What else did you learn?
c. What problems did you encounter and how did you solve them?

[Submit the completed lab report. Attach *Picture* of your work with all connections as part of your lab report]